MatLab for Engineers

This course is based on curriculum developed by:

- Nick Langhoff <langhoffn@smccd.edu>
- Erik Dunmire <edunmire@marin.edu>
- Tom Rebold <trebold@mps.edu>.

The course will be adapted and taught at Butte College by Evan Drake.

Course Description

This course utilizes the MATLAB environment and commercial electronic spreadsheets to provide students with a working knowledge of computer-based problem-solving methods relevant to science and engineering. It introduces the fundamentals of procedural and object-oriented programming, numerical analysis, and data structures. Examples and assignments in the course are drawn from practical applications in engineering, physics, and mathematics. (C-ID ENGR 220).

Course Objectives

Course objectives: There are essentially four major overlapping objectives in the course:

1. Introduce the MATLAB software environment.
2. Reinforce a structured, top-down approach to formulate and solve problems.
3. Introduce common approaches, structures, and conventions for creating and evaluating

computer programs, primarily in a procedural paradigm, but with a brief introduction to object-oriented concepts and terminology.
4. Apply a variety of common numeric techniques to solve and visualize engineering-related

computational problems. (Above objectives have been taken and reordered from original curriculum)

Teacher Information

email: <drakeev@butte.edu> Office Hours: M/W 10-11am, always at the butte college campus coffee shop.

Outline

This is a guide to the class and it is possible that things will be moved around as the class progresses

Week One

- Overview of course
- Overview of Matlab interface

Week Two

- Arrays
- Linear Algebra

Week Three

- Problem solving methods
- Files and Functions (fxn)

Week Four

- Polynomials & Data Structures
- Numerical Integrals & Derivatives

Week Five

- Logical Expressions
- Switch, Menu Structures

Week Six
- For loops
- Series Approximation

Week Seven
- While loop & Rootfinding
- Numerical Solution to ODEs

Week Eight
- Nested for loops, Vectorization, & 3D Plotting
- Logical Vectors

Week Nine
- Program Organization & Debugging
- Overview of Heat Transfer

Week Ten
- Stochastic Simulation

Week Eleven
- Strings
- Cell Arrays and Structures

Week Twelve
- Sorting & searching
- Formatted I/O to files

Week Thirteen
- More Advanced Functions
- OOP, Graphical Objects

Week fourteen
- Graphical User Interfaces (GUI)

Class Policy
- Attending both class is paramount to the success of students and highly recommended.
- I will do my best to provide extra material to study both course work and surrounding area, this is OPTIONAL.
- I have a zero late work policy

Butte College resources / Rules
- Academic guidelines
- Academic integrity
- Academic accommodations

Academic resources
- Center for academic success
- library website

Student technical resources
- Office 365 software
- Technical quickstart for students

- Canvas student guidelines